arXiv Corpus & Hybrid Retrieval

Phase 1 | Corpus statistics and implementation summary

Prepared 2026-11-04 11:22 UTC | Category: cs.CL

400 unique source papers across four subtopics, with **1200 text variants**. Variants are excerpts of the same sources and are not counted as additional papers.

Subtopic	Unique papers	Share
Machine Translation	100	25.0%
Question Answering	100	25.0%
Summarization	100	25.0%
Dialogue Systems	100	25.0%

Document lengths

Variant	Source PDF pages	Mean words	Median words
Short	5	2,987	3,032
Medium	8	4,753	4,850
Long	>9 (full)	11,639	10,308

Original PDF pages: min 10, max 208, mean 22.0, median 19.0. Publication years: 2025: 9, 2026: 391.

Collection and labeling

The arXiv Atom API was queried for cs.CL papers submitted from 2023-01-01 00:00 UTC to 2026-10-31 23:59 UTC. Papers were sorted by submission date descending. The collector accepted 100 globally unique, extractable papers of at least 10 pages for each topic. Overlapping matches were assigned to the first topic visited. Labels are keyword-derived; they have not been manually validated.

The short variant contains source pages 1-5; medium contains pages 1-8; long contains all pages. This gives controlled length comparisons but favors long source papers. These are page excerpts, not summaries. Confirm this interpretation of variants with the instructor.

Reproducibility & retrieval design

Exact arXiv queries

Machine Translation
cat:cs.CL AND all:"machine translation" AND submittedDate:[202301010000 TO 202610312359]

Question Answering
cat:cs.CL AND all:"question answering" AND submittedDate:[202301010000 TO 202610312359]

Summarization
cat:cs.CL AND all:summarization AND submittedDate:[202301010000 TO 202610312359]

Dialogue Systems
cat:cs.CL AND all:("dialogue system" OR "dialogue systems" OR "dialog system") AND submittedDate:[202301010000 TO 202610312359]

Snapshot evidence

data/manifest.json records the fixed date bounds, target, actual counts and collection timestamps. data/raw_api/ retains API responses. data/papers.jsonl records versioned arXiv URLs, original page counts, PDF checksums, variant paths, word counts and text checksums. data/rejected.jsonl logs short or text-poor sources. API ordering can change as records are revised; preserved responses and versioned URLs anchor this snapshot.

Retrieval and fusion

BM25 uses k1=1.5, b=0.75 and positive Robertson IDF over full variant text plus title. Dense retrieval uses pretrained all-MiniLM-L6-v2, attention-masked mean pooling, L2 normalization and cosine similarity. Full text is split into 120-word passages with 100-word stride and a title prefix. Inputs are limited to 256 model tokens. The maximum passage score becomes the document score.

Default fusion is RRF: sum 1/(60+rank) over sparse and dense ranks, with one-based ranks and sparse contributions only for lexical matches. Weighted fusion combines min-max normalized sparse and dense scores with selectable alpha. Both produce one fused list. Scores are ranking values, not probabilities.

Casefolding, abbreviation expansion, visible optional spelling correction and required quoted-phrase matching are implemented. The GUI provides ranking/variant selectors and top-10 unique-paper results with titles, links, approximately 200-character snippets and scores. Optional top-50 reranking adds a small title-term coverage bonus; it is not a trained reranker.

Verification and limitations

Corpus integrity audit: PASSED; 400 unique sources; 1200 checksum-verified text files; 400 local PDFs. All 12 mode/variant search combinations and query-handling integration checks passed.

Functional tests do not establish retrieval relevance. No human relevance judgments, precision/recall or effectiveness improvements are claimed. Other limitations: noisy PDF extraction, keyword-label ambiguity, long-source selection bias, possible input truncation, rare-name spelling errors, and max-passage scoring favoring documents with more passages. Original papers retain their individual rights. AI assistance is disclosed under ai-logs/phase1/.